

MATRIX

Problem

Given a matrix of size $N \times M$, print all elements of the matrix in spiral order starting from the top-left corner.

The traversal order is:

Left to Right across the top row
Top to Bottom along the right column
Right to Left across the bottom row
Bottom to Top along the left column

Continue this process until all elements are visited

Input Format

The first line contains two integers N and M , representing the number of rows and columns. The next N lines contain M space-separated integers representing the matrix.

Constraints

$1 \leq N, M \leq 1000$ $-10^9 \leq \text{Matrix}[i][j] \leq 10^9$

Output Format

Print the elements of the matrix in spiral order separated by spaces.

Sample Input 0

```
3 3
1 2 3
4 5 6
7 8 9
```

Sample Output 0

```
1 2 3 6 9 8 7 4 5
```

Sample Input 1

```
4 4
1 2 3 4
5 6 7 8
9 10 11 12
13 14 15 16
```

Sample Output 1

```
1 2 3 4 8 12 16 15 14 13 9 5 6 7 11 10
```

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Language Python 3



```
1 # Enter your code here. Read input from STDIN.
2 n, m = map(int, input().split())
3 a = []
4 for i in range(n):
5     a.append(list(map(int, input().split())))
6 top=left=0
7 bottom = n - 1
8 right = m - 1
9 ans = []
10 while top <= bottom and left <= right:
11     for j in range(left, right + 1):
12         ans.append(a[top][j])
13     top += 1
14     for i in range(top, bottom + 1):
15         ans.append(a[i][right])
16     right -= 1
17     if top <= bottom:
18         for j in range(right, left - 1, -1):
19             ans.append(a[bottom][j])
20     bottom -= 1
21     if left <= right:
22         for i in range(bottom, top - 1, -1):
23             ans.append(a[i][left])
24     left += 1
25 print(*ans)
```

Line: 6 Col: 5

Upload Code as File

Run Code

Submit Code

☐ Test against custom input

Congratulations

You solved this challenge. Would you like to challenge your friends?

Test case 0

Test case 1

Compiler Message

Success